

Unit 11 Test Study Guide

(Radical Expressions & Equations)

Name: Key!
Date: _____ Per: _____

Topic 1: Simplifying Radicals

Perfect Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225
Perfect Cubes: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, 1331, 1728, ...

1. $5\sqrt{112}$ $5\sqrt{16} \sqrt{7}$ $20\sqrt{7}$	2. $-\sqrt{44a^7b^{16}}$ $-\sqrt{4a^6b^{16}} \sqrt{11a}$ $-2a^3b^8\sqrt{11a}$	3. $2\sqrt{108x^9y^2z^{14}}$ $2\sqrt{36x^8y^2z^{14}} \sqrt{3x}$ $12x^4yz^7\sqrt{3x}$
4. $\sqrt[3]{189}$ $\sqrt[3]{27} \sqrt[3]{7}$ $3\sqrt[3]{7}$	5. $\sqrt[3]{-16a^4}$ $\sqrt[3]{-8a^3} \sqrt[3]{2a}$ $-2a\sqrt[3]{2a}$	6. $\sqrt[3]{320m^{15}n^7}$ $\sqrt[3]{64m^{15}n^6} \sqrt[3]{5n}$ $4m^5n^2\sqrt[3]{5n}$

Topic 2: Adding & Subtracting Radicals

7. $5\sqrt{7} + 2\sqrt{7} = 7\sqrt{7}$	8. $\sqrt{72} - \sqrt{18}$ $\sqrt{36} \sqrt{2} - \sqrt{9} \sqrt{2}$ $6\sqrt{2} - 3\sqrt{2} = 3\sqrt{2}$
9. $4\sqrt{48} - \sqrt{147} - 3\sqrt{50}$ $4\sqrt{16} \sqrt{3} - \sqrt{49} \sqrt{3} - 3\sqrt{25} \sqrt{2}$ $16\sqrt{3} - 7\sqrt{3} - 15\sqrt{2}$ $9\sqrt{3} - 15\sqrt{2}$	10. $3\sqrt{5} + 7\sqrt{125} - 2\sqrt{45}$ $3\sqrt{5} + 7\sqrt{25} \sqrt{5} - 2\sqrt{9} \sqrt{5}$ $3\sqrt{5} + 35\sqrt{5} - 6\sqrt{5}$ $32\sqrt{5}$

Topic 3: Multiplying Radicals

11. $5\sqrt{6} \cdot \sqrt{27}$ $5\sqrt{162}$ $5\sqrt{81} \sqrt{2}$ $45\sqrt{2}$	12. $7\sqrt{10} \cdot -3\sqrt{18}$ $-21\sqrt{180}$ $-21\sqrt{36} \sqrt{5}$ $-126\sqrt{5}$	13. $\sqrt{5x^2} \cdot \sqrt{8x^3}$ $\sqrt{40x^5}$ $\sqrt{4x^4} \sqrt{10x}$ $2x^2\sqrt{10x}$
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14. $\sqrt{6}(7\sqrt{3}+6)$

$$7\sqrt{18} + 6\sqrt{6}$$

$$7\sqrt{9}\sqrt{2} + 6\sqrt{6}$$

$$\boxed{21\sqrt{2} + 6\sqrt{6}}$$

15. $3\sqrt{2}(5\sqrt{10}-2\sqrt{6})$

$$15\sqrt{20} - 6\sqrt{12}$$

$$15\sqrt{4}\sqrt{5} - 6\sqrt{4}\sqrt{3}$$

$$\boxed{30\sqrt{5} - 12\sqrt{3}}$$

16. $(5+2\sqrt{7})(4-\sqrt{7})$

$$20 - 5\sqrt{7} + 8\sqrt{7} - 2\sqrt{49}$$

$$20 + 3\sqrt{7} - 14$$

$$\boxed{6 + 3\sqrt{7}}$$

17. $(1+2\sqrt{8})^2$

$$(1+2\sqrt{8})(1+2\sqrt{8})$$

$$1 + 2\sqrt{8} + 2\sqrt{8} + 4\sqrt{64}$$

$$1 + 4\sqrt{8} + 32$$

$$\boxed{33 + 4\sqrt{8}}$$

$$\boxed{33 + 8\sqrt{2}}$$

42

Topic 4: Dividing Radicals

18. $\sqrt{\frac{48}{3}} = \sqrt{16}$

$$= \boxed{4}$$

19. $\frac{\sqrt{18}}{\sqrt{75}} = \frac{\sqrt{6}}{\sqrt{25}}$

$$= \boxed{\frac{\sqrt{6}}{5}}$$

20. $\frac{6\sqrt{56}}{2\sqrt{63}} = \frac{3\cancel{6}\sqrt{8}}{2\sqrt{9}}$

$$= \frac{3\sqrt{4}\sqrt{2}}{2}$$

$$= \boxed{2\sqrt{2}}$$

21. $\sqrt{\frac{12}{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}}$

$$\frac{\sqrt{60}}{\sqrt{25}} = \frac{\sqrt{4}\sqrt{15}}{5}$$

$$= \boxed{\frac{2\sqrt{15}}{5}}$$

22. $\frac{12 \cdot \sqrt{2}}{3 \cdot 8\sqrt{2} \cdot \sqrt{2}}$

$$= \frac{\sqrt{2}}{3\sqrt{4}}$$

$$= \boxed{\frac{\sqrt{2}}{6}}$$

23. $\frac{5 \cdot 28\sqrt{8} \cdot \sqrt{3}}{4\sqrt{3} \cdot \sqrt{3}}$

$$\frac{5\sqrt{24}}{\sqrt{9}} = \frac{5\sqrt{4}\sqrt{6}}{3}$$

$$= \boxed{\frac{10\sqrt{6}}{3}}$$

24. $\frac{(5-\sqrt{3}) \cdot \sqrt{3}}{\sqrt{12} \cdot \sqrt{3}}$

$$\frac{5\sqrt{3} - \sqrt{9}}{\sqrt{36}}$$

$$= \boxed{\frac{-3 + 5\sqrt{3}}{6}}$$

25. $\frac{(\sqrt{8})(4-\sqrt{6})}{(4+\sqrt{6})(4-\sqrt{6})}$

$$\frac{4\sqrt{8} - \sqrt{48}}{16 - 4\sqrt{6} + 4\sqrt{6} - \sqrt{36}} = \frac{4\sqrt{2}\sqrt{4} - \sqrt{16}\sqrt{3}}{16 - 6}$$

$$= \frac{8\sqrt{2} - 4\sqrt{3}}{10}$$

$$= \boxed{\frac{4\sqrt{2} - 2\sqrt{3}}{5}}$$

Topic 5: Solving Radical Equations

26. $\frac{-2\sqrt{x}}{-2} = \frac{-16}{-2}$

$$\sqrt{x} = 8$$

$$\sqrt{x}^2 = 8^2$$

$$\boxed{x = 64}$$

CK

$$-2\sqrt{64} = -16$$

$$-2 \cdot 8 = -16$$

$$-16 = -16 \checkmark$$

27. $\frac{\sqrt{4x+1}+6}{-6-6} = \frac{11}{-6-6}$

$$\sqrt{4x+1} = 5$$

$$\sqrt{4x+1}^2 = 5^2$$

$$4x+1 = 25$$

$$4x = 24$$

$$\boxed{x = 6}$$

CK

$$\sqrt{4 \cdot 6 + 1} + 6 = 11$$

$$\sqrt{25} + 6 = 11$$

$$5 + 6 = 11$$

$$11 = 11 \checkmark$$

28. $x = \sqrt{56-x}$

$$x^2 = 56-x$$

$$x^2 + x - 56 = 0$$

$$(x+8)(x-7) = 0$$

$$x+8=0$$

$$\cancel{x = -8}$$

$$x-7=0$$

$$\boxed{x = +7}$$

$$-8 = \sqrt{56+8}$$

$$-8 = \sqrt{64}$$

$$-8 \neq 8$$

$$7 = \sqrt{56-7}$$

$$7 = \sqrt{49}$$

$$7 = 7 \checkmark$$

29. $p-3 = \sqrt{5p-15}$

$$(p-3)^2 = 5p-15$$

$$p^2 - 6p + 9 = 5p - 15$$

$$p^2 - 11p + 24 = 0$$

$$(p-8)(p-3) = 0$$

$$p-8=0$$

$$\boxed{p = 8}$$

$$p-3=0$$

$$\boxed{p = 3}$$

$$8-3 = \sqrt{5 \cdot 8 - 15}$$

$$5 = \sqrt{25}$$

$$5 = 5 \checkmark$$

$$3-3 = \sqrt{5 \cdot 3 - 15}$$

$$0 = \sqrt{0}$$

$$0 = 0 \checkmark$$